

PhysioGMC: Generalizable Multi-modal Coordination for Physiological Signals

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Abstract

Physiological signals are widely used for continuous health assessment in clinical and daily-life settings. Established physiological signals collected for inpatient and clinical use are often impractical for patients’ daily use due to their complexity and resource demands. In contrast, wearable signals enable continuous daily-life monitoring, but many have reduced reliability and limited clinical practices. To leverage complementary strengths of clinical and wearable physiological signals, we propose **PhysioGMC**, a **Generalizable Multi-modal Coordination** framework for **Physiological** signals that explicitly accounts for their strong inter-subject variability. PhysioGMC incorporates both clinical and wearable modalities into the training process to improve cross-subject performance when only a single wearable modality is available at deployment. The framework introduces a cross-modal contrastive learning module consisting of two contrastive losses to jointly learn label-relevant, subject-agnostic representations across modalities. Specifically, the self-supervised contrastive loss aligns latent features across modalities, while the supervised contrastive loss encourages learning label-discriminative features that are invariant to subject identity. Experiments on cardiovascular health monitoring and sleep staging tasks demonstrate that PhysioGMC consistently outperforms existing methods, achieving superior cross-subject performance using only wearable modalities, such as photoplethysmography (PPG), at test time.

1 Introduction

Continuous monitoring of physiological signals plays an essential role in assessing human health in both clinical and daily-life settings. In clinical practice, signals such as electrocardiography (ECG), arterial blood pressure (ABP), and polysomnography (PSG) are routinely collected for tasks including arrhythmia detection, hypertension management, and sleep staging. With the widespread adoption of wearable devices such as smartwatches, wearable signals, such as PPG

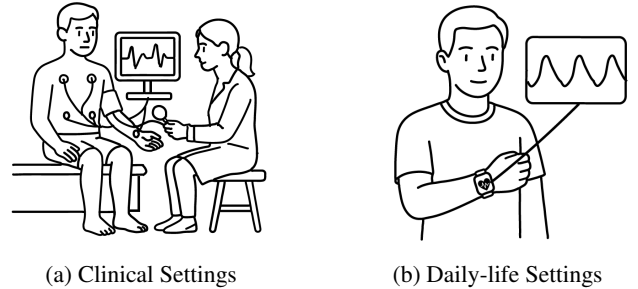


Figure 1: (a) An illustration of measuring ECG signals and blood pressure in clinical settings. (b) An illustration of continuous PPG signal monitoring via wearable devices in daily-life settings.

[Hwang *et al.*, 2021], have emerged as scalable, non-invasive alternatives for continuous health monitoring in daily life.

The growing volume of physiological data generated in and out of clinical settings has made manual expert review increasingly impractical, motivating the development of automated analysis systems using deep learning. Recent advances in those systems have shown remarkable success in health monitoring tasks such as atrial fibrillation (AF) detection [Torres-Soto and Ashley, 2020; Das *et al.*, 2022], hypertension assessment [Leitner *et al.*, 2022; BN *et al.*, 2024], sleep apnea detection [Olsen *et al.*, 2024; Sharan *et al.*, 2020], and sleep staging [Kotzen *et al.*, 2023]. Despite these successes, most existing approaches focus on a single physiological modality, overlooking the inherently complementary and interdependent nature of multi-modal physiological signals [Gil *et al.*, 2010; Mukkamala and Hahn, 2018].

Physiological signals collected in clinical settings, such as ECG, ABP, and PSG, provide more reliable, accurate, and clinically interpretable measurements of human health compared to signals measured by wearable devices. However, measuring these signals requires specialized and cumbersome equipment, making long-term continuous monitoring in non-clinical environments impractical, uncomfortable, and costly, as illustrated in Fig. 1a. In contrast, wearable devices enable more feasible and user-friendly long-term health monitoring through non-invasive measurements, as shown in Fig. 1b. This shows a trade-off between the fidelity of clinical-grade signals and the practicality of continuous real-world monitor-

ing. To balance this trade-off, it is desirable to complement the strengths of multi-modal physiological modalities by utilizing the inherent relationships among them to enhance the utility of wearable signals in real-world application scenarios.

Although training with multi-modal data has been shown to improve the performance of single-modality models [Zadeh *et al.*, 2020; Ma *et al.*, 2021; Liu *et al.*, 2023; Wu *et al.*, 2024], this paradigm is understudied in the context of physiological sensing. Existing approaches in this domain generally fall into two categories: 1) inferring the absent modalities during testing from the available ones; 2) aligning features across modalities to achieve multi-modal coordination. Inference-based methods [Li *et al.*, 2024; Vo *et al.*, 2024; Ji and Zhou, 2024] propose reconstructing absent physiological modalities from available modalities using generative models and subsequently applying the inferred signals to downstream tasks. However, such approaches are sensitive to reconstruction errors and may introduce additional noise that degrades performance. Recently, some work has focused on achieving multi-modal coordination for physiological signals. For example, [Guo *et al.*, 2023] employs a Siamese network to encourage learning shared features between ECG and PPG signals, improving AF detection accuracy using only PPG at inference. However, these existing methods overlook the substantial cross-subject differences inherent in physiological signals, hindering their generalization to unseen subjects.

In this work, we propose PhysioGMC, a multi-modal coordination framework designed for physiological signals, which coordinates wearable modalities with clinical-grade modalities during training to improve cross-subject performance of a single wearable modality at test time. During training, PhysioGMC employs a cross-modal contrastive learning module composed of two contrastive losses to align subject-agnostic, label-decisive features across clinical-grade and wearable-grade physiological modalities. The self-supervised cross-modal contrastive loss encourages consistency across features learned from different modalities of the same data instance, while the supervised cross-modal contrastive loss aims to learn label-decisive features agnostic to subjects. At test time, PhysioGMC achieves superior performance using only wearable signals, PPG, bridging the gap between the advantages of multi-modal learning and the constraints of real-world deployment scenarios with uni-modal wearable modalities.

To the best of our knowledge, we introduce the first generalizable multi-modal coordination framework for physiological signals. Extensive experiments on multiple public datasets demonstrate that our method consistently outperforms existing approaches across several health monitoring tasks, achieving superior cross-subject performance using only a wearable physiological signal, PPG, during inference.

2 Related Work

2.1 Inter-subject Variability in Physio. Signals

Physiological signals exhibit strong inter-subject variability, which limits the generalizability of models trained on a fixed set of subjects when deployed on unseen individuals. Hence, recent research has focused on learning subject-invariant features from physiological signals. For example, [Hwang *et*

al., 2021; Han *et al.*, 2024] adopts adversarial training to disentangle task-related, subject-invariant features from subject-specific features and noises, thereby mitigating inter-subject variability. [Kim *et al.*, 2024] propose leveraging both deep metric learning and contrastive learning strategies to learn task-related and subject-invariant features. While these approaches improve inter-subject generalization, they primarily operate in uni-modal settings and do not exploit complementary information across multiple physiological modalities.

2.2 Multi-modal Coordination for Physio. Signals

Previous studies [Zadeh *et al.*, 2020; Rahate *et al.*, 2022] have shown that models trained collaboratively using multi-modal data, which is referred to as multi-modal co-learning, can outperform those trained solely on uni-modal data in downstream uni-modal tasks. Motivated by these works, researchers explore multi-modal co-learning for physiological signals. [Liu *et al.*, 2024] proposes a cross-modal contrastive learning strategy to align ECG features with text features during the pretraining of an ECG-based encoder. [Ma *et al.*, 2025] improves sleep staging performance when only one modality is present at the testing stage by introducing modality-level contrastive coordination mechanisms. However, existing multi-modal co-learning approaches largely overlook the inter-subject variability intrinsic to physiological signals, which limits their effectiveness.

2.3 Multi-modal Domain Generalization (DG)

Most existing research for DG focuses on learning domain-invariant representations for uni-modal data. RNA-Net [Planamente *et al.*, 2022] is among the first methods to learn generalized multi-modal features across domains. [Dong *et al.*, 2023] aims to achieve domain generalization in multi-modal settings by splitting the representations into modality-specific and modality-shared components. [Fan *et al.*, 2024] proposes learning representations generalizable to unseen domains by constructing modality-consistent flat loss regions via cross-modal knowledge transfer.

3 Feasibility of Multi-modal Coordination for Physiological Signals

The practice of multi-modal learning paradigms has demonstrated that models trained with multiple modalities can outperform those trained on a single modality, even if only one modality is available at deployment. Previous studies [Zadeh *et al.*, 2020; Lu, 2023] attribute this advantage to the inter-related and complementary properties of the modalities used during training. Building on this insight, we extend the theoretical analysis in [Zadeh *et al.*, 2020] to formalize the benefits of multi-modal coordination over uni-modal learning.

Theorem 1. *Let $\mathcal{F}_\Theta$ denote a model with learnable parameters Θ . During training, the model has access to multi-modal inputs $\{x_i, y \mid i \in M\}$, where i indexes modalities within the full set of modalities M , and y denotes labels for downstream tasks. At test time, only a subset of modalities is available, denoted as $\{x_m \mid m \in S, S \subset M\}$, where m indexes the modalities in the subset available at test time S . The complementary set of S in M is defined as $S^c = M \setminus S$.*

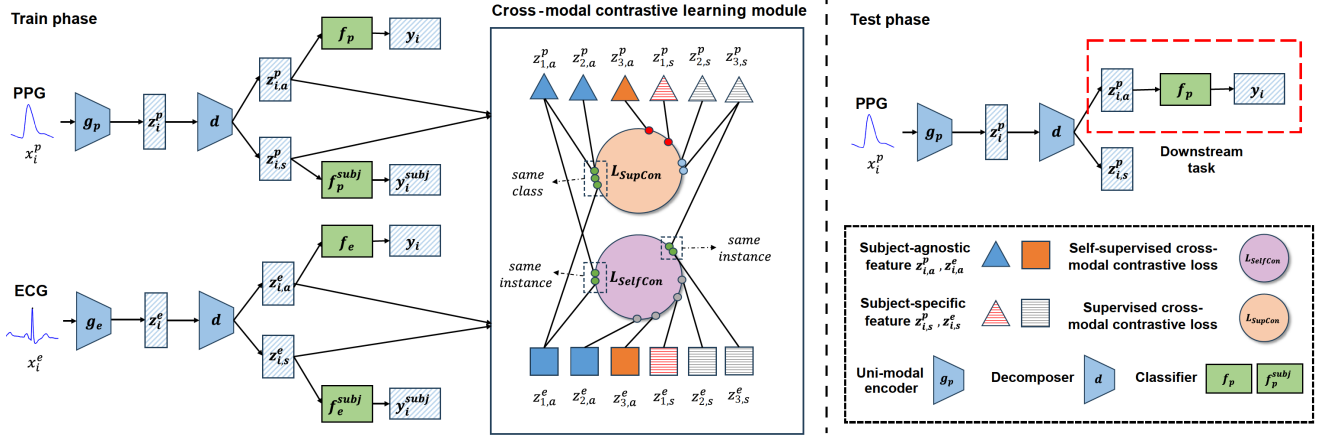


Figure 2: The overall illustration of the proposed framework. Here, we take the example of using PPG and ECG signals as input during the training phase. The color indicates categories, while the shape indicates modality types.

Assume that the modalities in M are interrelated with each other ($I(\{x_m | m \in S\}; \{x_n | n \in S^c\}) > 0$) and that incorporating more modalities enhances the downstream tasks in terms of $H(y | \{x_m | m \in M\}) < H(y | \{x_m | m \in S\}) < H(y)$, where $I(\cdot; \cdot)$ denotes mutual information and $H(\cdot)$ denotes entropy. Under these assumptions, the following relationship holds:

$$\exists \theta \in \Theta, H(y | \{x_m | m \in M\}) < H(y | \{x_m | m \in S\}, \theta) < H(y | \{x_m | m \in S\}) \quad (1)$$

For the θ in Eq. 1, the following relationships must hold,

$$H(\{x_n | n \in S^c\} | \{x_m | m \in S\}) > H(\{x_n | n \in S^c\} | \{x_m | m \in S\}, \theta) \quad (2)$$

$$I(y; \{x_n | n \in S^c\} | \{x_m | m \in S\}) > I(y; \{x_n | n \in S^c\} | \{x_m | m \in S\}, \theta) \quad (3)$$

Theorem 1 demonstrates that, by capturing connections between modality sets $\{x_m | m \in S\}$ and $\{x_n | n \in S^c\}$ that preserve label-relevant information, as illustrated in Eq. 2 and 3, multi-modal models can outperform uni-modal counterparts, even when only a subset of modalities is available at inference, as stated in Eq. 1.

To empirically validate this principle in the context of physiological signals, we adopt multi-modal SimCLR [Chen *et al.*, 2020] as a representative multi-modal co-learning framework and apply it to the AF detection task. It enforces cross-modal alignment by treating temporally synchronized signal samples from different modalities as positive pairs, encouraging the learned label-decisive representations to encode shared information across modalities. Therefore, SimCLR instantiates the conditions for effective multi-modal co-learning formalized in Theorem 1, making it an appropriate choice for empirically validating the benefits of multi-modal co-learning over uni-modal learning.

Despite improved performance over uni-modal baselines, results in Table 1 show that conventional multi-modal coordination methods suffer from notable inter-subject variability

Table 1: The performance comparison of the AF detection task on a subset of MIMIC-III under intra-subject and inter-subject test scenarios. The intra-subject evaluation uses data from the same set of subjects, with training and testing conducted on samples collected at different times. In the inter-subject setting, performance is evaluated on data from subjects excluded during training.

	MIMIC-III [Johnson <i>et al.</i> , 2016]			
	ECG		PPG	
	Accuracy($\uparrow$)	Macro F1($\uparrow$)	Accuracy($\uparrow$)	Macro F1($\uparrow$)
Intra-subject eval.				
Uni-modal	0.980	0.980	0.858	0.857
Multi-modal SimCLR	0.988	0.988	0.880	0.879
Inter-subject eval.				
Uni-modal	0.672	0.671	0.651	0.651
Multi-modal SimCLR	0.680	0.679	0.683	0.683

and limited generalization to unseen subjects, motivating our proposed framework that learns generalizable representations collaboratively from multi-modal physiological signals.

4 Method

An overview of the proposed framework is shown in Fig. 2. It comprises two primary components: 1) Uni-modal feature extraction and classification: this component serves as the backbone of PhysioGMC to extract features from each physiological modality; 2) Cross-modal contrastive learning module: This module jointly uses a self-supervised contrastive loss to align instance-level features across modalities and a supervised contrastive loss to align label-relevant features of the same class across modalities while separating them from features of other classes and from subject-specific features. These two cross-modal contrastive learning objectives enable effective and generalizable multi-modal coordination across physiological modalities.

4.1 Uni-modal Feature Extraction and Classification

In the proposed framework, we adopt EfficientNet [Tan and Le, 2019] as the backbone encoder for uni-modal feature ex-

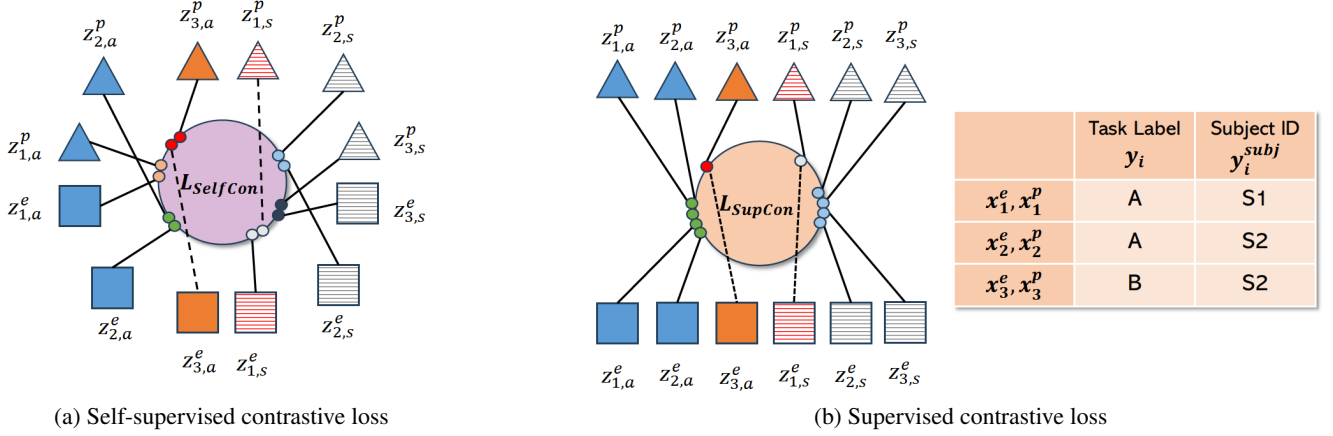


Figure 3: (a) An illustration of the proposed self-supervised cross-modal contrastive loss. (b) An illustration of the proposed supervised cross-modal contrastive loss. The legends of two sub-figures are the same as Fig. 2. As introduced in Section 4.1, $(z_{i,a}^m, z_{i,s}^m) = d(g_m(x_i^m))$, where $z_{i,a}^m$ and $z_{i,s}^m$ denote the label-relevant features and subject-specific features, respectively, from the input data instance x_i^m of modality m . Along the circles, features pointing to dots of the same color are treated as positive pairs for the contrastive loss. The table in Fig. 3b illustrates the label information for sample input data instances of this figure.

traction. For a given data instance of the modality m , denoted as x_i^m , the corresponding uni-modal feature is obtained as $z_i^m = g_m(x_i^m)$, where $g_m(\cdot)$ is the modality-specific feature encoder and $i \in \{1, \dots, N\}$ is the data instance index within a batch. The learned feature z_i^m is further decomposed into the label-relevant, subject-agnostic feature $z_{i,a}^m$ and the subject-specific feature $z_{i,s}^m$ using a feature decomposer $d(\cdot)$: $d(z_i^m) = [z_{i,a}^m; z_{i,s}^m]$. These decomposed features are subsequently used for downstream classification.

The modality-specific classification loss is defined as:

$$\mathcal{L}_{cls}^m = \frac{1}{N} \sum_{i=1}^N (\mathcal{L}_{CE}(f_m(z_{i,a}^m), y_i) + \mathcal{L}_{CE}(f_m^{subj}(z_{i,s}^m), y_i^{subj})), \quad (4)$$

where $\mathcal{L}_{CE}$ denotes the cross-entropy (CE) loss, f_m and f_m^{subj} are the modality-specific classifiers, y_i is the label of downstream tasks and y_i^{subj} is the subject ID label. Minimizing the above classification loss encourages the model to learn the label-related features and the identity-related features for each modality separately.

4.2 Cross-modal Contrastive Learning

As discussed in Section 3, multi-modal learning paradigms can outperform uni-modal counterparts when they capture label-decisive mappings across different modalities. Motivated by this insight, we propose a cross-modal contrastive learning module that is composed of two contrastive losses: self-supervised contrastive loss and supervised contrastive loss. This module is designed to learn subject-agnostic, label-relevant features that are consistently aligned across modalities, thereby implicitly modeling label-decisive relationships between different physiological modalities.

For the contrastive learning, the extracted uni-modal features of the modality m are normalized using L_2 normalization as $\tilde{z}_{i,a}^m = \frac{z_{i,a}^m}{\|z_{i,a}^m\|}$, $\tilde{z}_{i,s}^m = \frac{z_{i,s}^m}{\|z_{i,s}^m\|}$. We assume that each

batch of the data used for training consists of N quadruplets, $\{\mathbf{x}_i, y_i, y_i^{subj}\}_{i=1, \dots, N}$, where $\mathbf{x}_i = [x_i^m; x_i^k]$ denotes temporally synchronized data instances from two physiological modalities m and k , and i is the data index.

Self-supervised Contrastive Loss

As the first component of the proposed module, we introduce a self-supervised cross-modal contrastive loss, defined in Eq. 5, which enforces alignment between representations from the same data instance across different modalities.

$$\mathcal{L}_{SelfCon} = -\frac{1}{N} \sum_{i=1}^N \left(\log \frac{\exp(\tilde{z}_{i,a}^m \cdot \tilde{z}_{i,a}^k / \tau)}{\sum_{j=1}^N \exp(\tilde{z}_{j,a}^m \cdot \tilde{z}_{j,a}^k / \tau)} + \log \frac{\exp(\tilde{z}_{i,s}^m \cdot \tilde{z}_{i,s}^k / \tau)}{\sum_{j=1}^N \exp(\tilde{z}_{j,s}^m \cdot \tilde{z}_{j,s}^k / \tau)} \right), \quad (5)$$

where τ is the temperature scaling parameter, controlling the sharpness of the similarity scores, and $\cdot$ is the inner product of two normalized features for measuring their similarity.

The self-supervised contrastive loss encourages representations extracted from temporally paired modalities to be aligned in the embedding space, without relying on label supervision. Consequently, it enables the effective incorporation of large-scale unlabeled multi-modal physiological data during training. An illustration of the self-supervised cross-modal contrastive loss is provided in Fig. 3a.

Supervised Contrastive Loss

Although the self-supervised contrastive loss effectively aligns features across modalities, the learned features may still contain label-irrelevant or subject-specific information. The entanglement of these two kinds of information can hinder discrimination and generalization of learned representations in downstream tasks, leading to degraded cross-subject performance. In light of this, we propose a modified supervised contrastive loss to learn multi-modal features that only

retain subject-agnostic, label-decisive information. This contrastive loss utilizes both task and subject label information as supervision, drawing features from the same class closer in the embedding space while pushing apart features from different classes, as illustrated in Fig. 3b. This loss consists of two loss terms: the uni-modal contrastive loss term and the cross-modal contrastive loss term.

To construct the proposed supervised contrastive loss, we concatenate the normalized label-relevant subject-agnostic feature $\{\widetilde{z}_{i,a}^m\}_{i=1,\dots,N}$ and the normalized subject-specific feature $\{\widetilde{z}_{i,s}^m\}_{i=1,\dots,N}$ into a new feature vector $z'_m = [\{\widetilde{z}_{i,a}^m\}_{i=1,\dots,N}; \{\widetilde{z}_{i,s}^m\}_{i=1,\dots,N}]$. The task label y_i is assigned to $\widetilde{z}_{i,a}^m$ while the subject ID label y_i^{subj} is assigned to $\widetilde{z}_{i,s}^m$. Consequently, we define the combined label vector as $y' = \{y_i\}_{i=1,\dots,N}; \{y_i^{subj}\}_{i=1,\dots,N}$. The modified supervised contrastive loss is denoted as follows:

$$\begin{aligned} \mathcal{L}_{\text{SupCon}} &= \mathcal{L}_{\text{SupCon}}^{\text{uni}} + \beta_1 \mathcal{L}_{\text{SupCon}}^{\text{cross}} \\ &= - \sum_{m \in \mathcal{M}} \sum_{i=1}^{2N} \frac{1}{N'} \sum_{j \in P(i)} \log \frac{\exp(z'_{m,i} \cdot z'_{m,j} / \tau)}{\sum_{r \in A(i)} \exp(z'_{m,i} \cdot z'_{m,r} / \tau)} \\ &\quad - \beta_1 \sum_{m \in \mathcal{M}} \sum_{k \in \mathcal{M}^c} \sum_{i=1}^{2N} \frac{1}{N'} \sum_{j \in P(i)} \log \frac{\exp(z'_{m,i} \cdot z'_{k,j} / \tau)}{\sum_{r \in A(i)} \exp(z'_{m,i} \cdot z'_{k,r} / \tau)} \end{aligned} \quad (6)$$

where $i \in I \equiv \{1, \dots, 2N\}$ is the index of data instance within a batch. We define $P(i) \equiv \{p : y'_p = y'_i\}$ and $A(i) \equiv I \setminus \{i\}$. N' is the cardinality of $P(i)$. $\mathcal{M}$ is the set of available modalities during the training process. β_1 is the hyperparameter that controls the relative importance of uni-modal and cross-modal contrastive learning terms. τ and the similarity operator $\cdot$ have the same definitions as in the self-supervised contrastive loss.

In this supervised contrastive loss, the uni-modal contrastive loss term encourages within-class consistency by attracting the same-class features within each modality. Complementarily, the cross-modal contrastive loss term encourages inter-modal consistency by aligning same-class features across modalities. Together, these terms guide the model to extract label-relevant information from multi-modal data.

The supervised contrastive loss based solely on task labels learns representations that can still retain subject-specific information, limiting their generalization to unseen subjects. To mitigate this concern, we incorporate subject-specific features $\widetilde{z}_{i,s}^m$ labeled with subject identities into the modified supervised contrastive loss. By treating the label-relevant feature $\widetilde{z}_{i,a}^m$ and the subject-specific feature $\widetilde{z}_{i,s}^m$ extracted from the same data instance as belonging to different classes, the proposed supervised contrastive loss effectively disentangles label-relevant information and subject-specific information, yielding label-relevant representations that are more generalizable across subjects.

4.3 Final Loss

The final loss function is defined as the weighted sum of the previously defined losses:

$$\mathcal{L} = \lambda_{cls} \sum_{m \in \mathcal{M}} \mathcal{L}_{cls}^m + \lambda_{self} \mathcal{L}_{SelfCon} + \lambda_{sup} \mathcal{L}_{SupCon}, \quad (7)$$

where λ_{cls} , λ_{self} , and λ_{sup} are the hyperparameters that control the relative importance of different loss terms.

5 Experiments and Results

5.1 Experimental Settings

Datasets. We conduct experiments on two subsets derived from the real-world ICU dataset MIMIC-III [Johnson *et al.*, 2016], referred to as MIMIC-AF and MIMIC-Hyper, as well as on a subset of the MESA dataset [Chen *et al.*, 2015], which includes 2237 participants enrolled in a sleep exam. Both MIMIC-AF and MIMIC-Hyper consist of physiological recordings from 300 critically ill subjects. The former subset contains ECG and PPG signals, while the latter contains ABP and PPG signals. All signals are segmented into 10-second windows for both ICU subsets. The MESA subset consists of overnight PSG recordings from 100 subjects, where each recording contains synchronized 30-second ECG and PPG segments.

Downstream Tasks. We use two cardiovascular health monitoring tasks and one sleep monitoring task for evaluation over different datasets: atrial fibrillation (AF) detection on the MIMIC-AF, blood pressure classification on the MIMIC-Hyper, and sleep stage classification on the MESA subset. For the MIMIC-AF, each segment is labeled according to the corresponding ICD-9 diagnostic codes. For the MIMIC-Hyper, segments are categorized into four classes: normal, high normal, Grade 1 hypertension, and Grades 2–3 hypertension. The categorization is based on their systolic blood pressure (SBP) and diastolic blood pressure (DBP) values, following the ESC/ESH guidelines [Mancia *et al.*, 2014]. For the MESA subset, we consider a four-class sleep staging task comprising wake, light sleep (N1/N2), deep sleep (N3), and Rapid Eye Movement (REM).

Evaluation Settings. To evaluate the generalization performance of the proposed framework, we perform a five-fold cross-subject validation with no overlap between training and testing subjects. The final results are averaged across all five folds for an overall evaluation. We run each experiment with five random seeds to account for performance variance and report the mean and standard deviation of the five runs' results. Additionally, the Wilcoxon signed-rank test is used to assess the statistical significance of the performance improvements achieved by our proposed methods over the baseline methods.

5.2 Experiment Results

To evaluate the proposed method, we compare it with eight baseline methods. All methods are evaluated under a train-with-multi-modal, test-with-uni-modal setting, where ECG or ABP signals are available as auxiliary modalities only during training, while PPG serves as the primary modality available during both training and inference.

Table 2: The performance comparison of baseline methods and the proposed framework on MIMIC-AF and MIMIC-Hyper datasets for AF detection and blood pressure classification tasks when using multi-modal signals as input during the training stage and only using PPG signals as input during the inference stage. The bold and underlined items denote the best and second-best results, respectively. An asterisk (*) indicates that the proposed method achieves a significant improvement over the best baseline method, with a $p < 0.05$.

Method	MIMIC-AF			MIMIC-Hyper		
	Accuracy ($\uparrow$)	Macro F1 ($\uparrow$)	AUC.ROC ($\uparrow$)	Accuracy ($\uparrow$)	Macro F1 ($\uparrow$)	AUC.ROC ($\uparrow$)
Uni-modal						
Supervised w/ CE	0.6667 $\pm$ 0.003	0.6658 $\pm$ 0.003	0.7236 $\pm$ 0.004	0.4146 $\pm$ 0.006	0.3274 $\pm$ 0.002	0.6089 $\pm$ 0.003
Supervised w/ CE	0.667 $\pm$ 0.003	0.666 $\pm$ 0.003	0.724 $\pm$ 0.004	0.415 $\pm$ 0.006	0.327 $\pm$ 0.002	0.609 $\pm$ 0.003
Uni-modal DG						
BPD [Su <i>et al.</i> , 2022]	0.681 $\pm$ 0.003	0.679 $\pm$ 0.003	0.734 $\pm$ 0.010	0.410 $\pm$ 0.007	0.334 $\pm$ 0.003	<u>0.634</u> $\pm$ 0.008
CSGR [Kim <i>et al.</i> , 2024]	0.657 $\pm$ 0.004	0.655 $\pm$ 0.004	0.711 $\pm$ 0.007	0.414 $\pm$ 0.005	0.324 $\pm$ 0.002	0.603 $\pm$ 0.002
Multi-modal						
Multi-modal SimCLR [Chen <i>et al.</i> , 2020]	<u>0.687</u> $\pm$ 0.004	<u>0.686</u> $\pm$ 0.004	<u>0.748</u> $\pm$ 0.007	<u>0.427</u> $\pm$ 0.005	<u>0.339</u> $\pm$ 0.003	0.624 $\pm$ 0.009
MERL [Liu <i>et al.</i> , 2024]	0.670 $\pm$ 0.007	0.668 $\pm$ 0.007	0.724 $\pm$ 0.005	0.415 $\pm$ 0.006	0.327 $\pm$ 0.003	0.612 $\pm$ 0.009
SiamAF [Guo <i>et al.</i> , 2023]	0.661 $\pm$ 0.004	0.661 $\pm$ 0.004	0.715 $\pm$ 0.006	0.422 $\pm$ 0.004	0.326 $\pm$ 0.003	0.608 $\pm$ 0.007
Multi-modal DG						
SimMMDG [Dong <i>et al.</i> , 2023]	0.667 $\pm$ 0.004	0.665 $\pm$ 0.004	0.720 $\pm$ 0.006	0.421 $\pm$ 0.004	0.325 $\pm$ 0.002	0.603 $\pm$ 0.005
CMRF [Fan <i>et al.</i> , 2024]	0.660 $\pm$ 0.002	0.659 $\pm$ 0.001	0.718 $\pm$ 0.002	0.417 $\pm$ 0.002	0.323 $\pm$ 0.003	0.607 $\pm$ 0.004
Proposed	0.713* $\pm$ 0.005	0.711* $\pm$ 0.005	0.776* $\pm$ 0.005	0.436* $\pm$ 0.005	0.360* $\pm$ 0.002	0.649* $\pm$ 0.004

Table 3: The performance comparison of baseline methods and the proposed framework on the MESA subset for the sleep stage classification tasks. The bold and underlined items denote the best and second-best results, respectively. The asterisk has the same meaning as Table. 2.

Method	MESA subset			
	Accuracy ($\uparrow$)	Macro F1 ($\uparrow$)	AUC.ROC ($\uparrow$)	Kappa ($\uparrow$)
Uni-modal				
Supervised w/ CE	0.493 $\pm$ 0.007	0.442 $\pm$ 0.005	0.720 $\pm$ 0.003	0.286 $\pm$ 0.006
Uni-modal DG				
BPD	0.488 $\pm$ 0.008	0.451 $\pm$ 0.005	0.729 $\pm$ 0.003	0.292 $\pm$ 0.007
CSGR	0.484 $\pm$ 0.008	0.442 $\pm$ 0.004	0.720 $\pm$ 0.005	0.286 $\pm$ 0.005
Multi-modal				
Multi-modal SimCLR	<u>0.504</u> $\pm$ 0.010	<u>0.457</u> $\pm$ 0.006	0.736 $\pm$ 0.004	<u>0.302</u> $\pm$ 0.010
MERL	0.492 $\pm$ 0.006	0.450 $\pm$ 0.003	0.726 $\pm$ 0.005	0.293 $\pm$ 0.006
SiamAF	0.464 $\pm$ 0.008	0.408 $\pm$ 0.009	0.673 $\pm$ 0.009	0.237 $\pm$ 0.011
Multi-modal DG				
SimMMDG	0.472 $\pm$ 0.037	0.395 $\pm$ 0.015	0.717 $\pm$ 0.011	0.259 $\pm$ 0.026
CMRF	0.449 $\pm$ 0.006	0.400 $\pm$ 0.005	0.734 $\pm$ 0.002	0.246 $\pm$ 0.003
Proposed	0.517* $\pm$ 0.004	0.461 $\pm$ 0.004	<u>0.735</u> $\pm$ 0.005	0.314* $\pm$ 0.004

Table 4: The experiment results of AF detection tasks with different label missing ratios (0%, 40%, 70%) for the MIMIC-AF dataset.

Method	Macro F1 ($\uparrow$)		
	0%	40%	70%
Supervised w/ CE	0.666 $\pm$ 0.003	0.661 $\pm$ 0.002	0.656 $\pm$ 0.002
Proposed	0.711 $\pm$ 0.005	0.708 $\pm$ 0.001	0.705 $\pm$ 0.004

The Cardiovascular Health Monitoring.

The experimental results for AF detection and blood pressure classification are summarized in Table 2. First, we observe that uni-modal methods designed to address inter-subject variability in physiological signals, such as BPD [Su *et al.*, 2022], achieve better cross-subject testing performance compared to the vanilla supervised learning counterpart. Second, certain multi-modal coordination methods, such as SimCLR [Chen *et al.*, 2020], outperform uni-modal counterparts, demonstrating the effectiveness of multi-modal coordination

methods when only uni-modal data is available during deployment. In contrast, the two multi-modal DG methods [Dong *et al.*, 2023; Fan *et al.*, 2024] show inferior performance, likely due to the strong dominance of ECG/ABP signals in their multi-modal framework, which may hinder the models’ generalization when only PPG signals are available at test time.

Overall, our proposed method consistently outperforms all baseline approaches across all three evaluation metrics on two cardiovascular health monitoring tasks. It achieves about 2.5% absolute improvement in *Accuracy*, *Macro F1*, and *AUC.ROC* scores for the AF detection task compared to the second-best method. For the blood pressure classification task, it improves *Accuracy* by 1%, *Macro F1* by 2%, and *AUC.ROC* scores by 1.6%. Specifically, the proposed method achieves statistically significant improvements over the second-best method across all evaluation metrics.

The Sleep Stage Classification.

Table 3 presents experimental results of the sleep stage classification task on the MESA subset. Consistent with the findings in the previous section, multi-modal coordination methods, including SimCLR [Chen *et al.*, 2020] and MERL [Liu *et al.*, 2024], outperform uni-modal approaches. The proposed framework, PhysioGMC, achieves the best performance among all evaluated methods. These results demonstrate the superiority of learning from various physiological signals collaboratively during the training process, even when only uni-modal wearable signals are available at deployment, which offers more practical and comfortable options for monitoring sleep continuously.

Extension to Semi-supervised Learning Settings

As discussed in the *Self-supervised Contrastive Loss* section of Section 4.2, the proposed framework can extend to semi-supervised learning settings where models are trained using both labeled data and unlabeled data. In this section, we evaluate the performance of PhysioGMC under the semi-

Table 5: Ablation study on the influence of each individual component for the MIMIC-AF and MIMIC-Hyper dataset. The bold and underlined items denote the best and second-best results, respectively. The asterisk has the same meaning as Table. 2.

ID Method	MIMIC-AF			MIMIC-Hyper		
	Accuracy ($\uparrow$)	Macro F1 ($\uparrow$)	AUC-ROC ($\uparrow$)	Accuracy ($\uparrow$)	Macro F1 ($\uparrow$)	AUC-ROC ($\uparrow$)
<i>Uni-modal</i>						
A1 Supervised w/ CE	0.667 $\pm$ 0.003	0.666 $\pm$ 0.003	0.724 $\pm$ 0.004	0.415 $\pm$ 0.006	0.327 $\pm$ 0.002	0.609 $\pm$ 0.003
A2 Supervised contrastive loss	0.681 $\pm$ 0.005	0.678 $\pm$ 0.005	0.728 $\pm$ 0.007	0.409 $\pm$ 0.011	0.337 $\pm$ 0.005	0.626 $\pm$ 0.007
<i>Multi-modal</i>						
B1 Proposed w/o self-supervised contrastive loss	0.682 $\pm$ 0.006	0.681 $\pm$ 0.006	0.735 $\pm$ 0.012	0.409 $\pm$ 0.011	0.334 $\pm$ 0.006	0.628 $\pm$ 0.007
B2 Proposed w/o supervised contrastive loss	0.682 $\pm$ 0.002	0.681 $\pm$ 0.002	0.738 $\pm$ 0.002	0.430 $\pm$ 0.005	0.347 $\pm$ 0.003	0.627 $\pm$ 0.005
— Proposed	0.713 $\pm$ 0.005	0.711 $\pm$ 0.005	0.776 $\pm$ 0.005	0.436 $\pm$ 0.005	0.359 $\pm$ 0.002	0.649 $\pm$ 0.004

supervised scenario by randomly masking labels at ratios of {0%, 40%, 70%}. We compare PhysioGMC with the uni-modal supervised learning baseline trained only on the available labeled data. As shown in Table 4, PhysioGMC outperforms the baseline across all label-missing ratios, demonstrating its scalability in semi-supervised settings by leveraging unlabeled data through the self-supervised cross-modal contrastive loss in the training process.

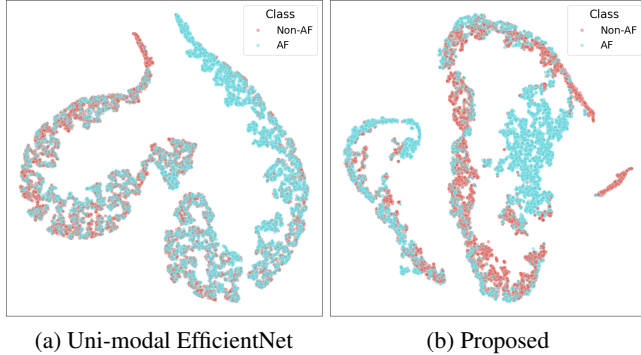


Figure 4: Visualization of the t-SNE embeddings of PPG signals in the MIMIC-AF dataset using (a) Uni-modal EfficientNet and (b) Proposed PhysioGMC framework.

5.3 Ablation Experiments

To further illustrate the contributions of each module in the proposed framework, we conduct ablation experiments on the MIMIC-AF and MIMIC-Hyper datasets. The results of ablation experiments, as shown in Table 5, demonstrate the effectiveness of both self-supervised and supervised cross-modal contrastive losses. The variant B2, which only incorporates the self-supervised contrastive loss, achieves notable improvements in performance compared to A1 under uni-modal testing scenarios, demonstrating the advantage of multi-modal coordination paradigms that encourage the learning of modality-consistent features. The proposed framework, which additionally incorporates the supervised contrastive loss, further enhances cross-subject performance over B2, as it explicitly enforces the disentanglement of subject-agnostic information and subject-specific information in the learned multi-modal label-decisive features.

We also evaluate the effectiveness of the proposed supervised contrastive loss when only a single modality is used

for training. Compared to the uni-modal baseline A1, variant A2 achieves improved cross-subject test performance, suggesting its ability to learn label-relevant representations agnostic to subjects. However, the proposed framework with multi-modal contrastive learning modules outperforms this uni-modal variant, demonstrating the advantage of learning subject-agnostic, label-relevant features collaboratively from multiple modalities rather than learning them from a single modality alone.

5.4 Feature Visualization Analysis

In this section, we utilize t-SNE [Maaten and Hinton, 2008] to visualize the embeddings of PPG signals learned by the baseline uni-modal model and PhysioGMC on the MIMIC-AF dataset. As shown in Fig. 4, the embeddings learned by PhysioGMC exhibit clearer and more distinguishable classification boundaries. For subjects without AF, the distribution of PPG embeddings is inherently more dispersed due to heterogeneous cardiovascular conditions. Nevertheless, the proposed framework yields a more compact distribution for the non-AF cohort compared to the uni-modal baseline, demonstrating its ability to suppress subject-specific information in the extracted label-relevant features. While there remains some overlap between classes, reflecting the challenges brought by inter-subject variability of PPG signals, these results show improved feature separability for our framework. Further exploration may help achieve more distinct decision boundaries for unseen subjects.

6 Conclusion

We propose PhysioGMC, a generalizable multi-modal coordination framework for physiological signals that leverages both clinical-grade and wearable modalities during training. The proposed framework learns label-relevant, subject-agnostic features collaboratively from multiple modalities through integration of proposed self-supervised and supervised cross-modal contrastive losses, leading to improved cross-subject performance of wearable modalities under uni-modal deployment scenarios. Experiments on cardiovascular health monitoring and sleep stage classification tasks show that PhysioGMC consistently achieves superior cross-subject performance when only a single wearable physiological signal, PPG, is available during testing, highlighting the potential of PhysioGMC to support continuous, accessible, and reliable health monitoring using only wearable sensing in real-world applications.

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A Appendices.

A.1 Proof for Theorem 1

Let $\mathcal{F}_\Theta$ denote a model parameterized by learnable parameters Θ . During training, the model has access to multimodal data $\{x_i \mid i \in M\}$ and the corresponding downstream-task label y , where M denotes the full set of modalities. At test time, only a subset of modalities, $\{x_m \mid m \in S\}$, is available, where $S \subset M$. The set of modalities unavailable at test time is given by the complement $S^c = M \setminus S$. We make the following two assumptions in this theorem, where $I(\cdot; \cdot)$ denotes mutual information and $H(\cdot)$ denotes entropy:

- 1) The full set of modalities is interrelated with each other $I(\{x_m \mid m \in S\}; \{x_n \mid n \in S^c\}) > 0$;
- 2) The incorporation of more modalities enhances the downstream tasks in terms of $H(y \mid \{x_m \mid m \in M\}) < H(y \mid \{x_m \mid m \in S\}) < H(y)$.

Below are the statements for Theorem 1:

$$\exists \theta \in \Theta, H(y \mid \{x_m \mid m \in M\}) < H(y \mid \{x_m \mid m \in S\}, \theta) < H(y \mid \{x_m \mid m \in S\}) \quad (8)$$

The first inequality, $H(y \mid \{x_m \mid m \in M\}) < H(y \mid \{x_m \mid m \in S\}, \theta)$, represents the lower bound of the entropy achievable for the multimodal co-learning, corresponding to the ideal scenario in which all modalities are available during both training and testing and $\{x_n \mid n \in S^c\}$ is not fully predictable from $\{x_m \mid m \in S\}$ using model θ :

$$H(y \mid \{x_m \mid m \in M\}) = H(y \mid \{x_m \mid m \in S\}, \{x_n \mid n \in S^c\}) < H(y \mid \{x_m \mid m \in S\}, \theta) \quad (9)$$

The following condition must be satisfied for the second inequality, $H(y \mid \{x_m \mid m \in S\}, \theta) < H(y \mid \{x_m \mid m \in S\})$, to hold:

$$I(y; \theta \mid \{x_m \mid m \in S\}) > 0 \quad (10)$$

To ensure that the above conditions are satisfied, the following expression must hold:

$$I(y; \theta \mid \{x_m \mid m \in S\}) = I(y; \theta, \{x_n \mid n \in S^c\} \mid \{x_m \mid m \in S\}) - I(y; \{x_n \mid n \in S^c\} \mid \{x_m \mid m \in S\}, \theta) > 0 \quad (11)$$

Also, we can derive the following equation:

$$I(y; \theta, \{x_n \mid n \in S^c\} \mid \{x_m \mid m \in S\}) = I(y; \{x_n \mid n \in S^c\} \mid \{x_m \mid m \in S\}) + I(y; \theta \mid \{x_n \mid n \in S^c\}, \{x_m \mid m \in S\}) \quad (12)$$

Given the assumption, $H(y \mid \{x_m \mid m \in S\}, \{x_n \mid n \in S^c\}) < H(y \mid \{x_m \mid m \in S\}) < H(y)$, equation 11 hold if the conditions below are satisfied:

$$I(y; \{x_n \mid n \in S^c\} \mid \{x_m \mid m \in S\}) > 0 \quad (13)$$

$$I(y; \{x_n \mid n \in S^c\} \mid \{x_m \mid m \in S\}) - I(y; \{x_n \mid n \in S^c\} \mid \{x_m \mid m \in S\}, \theta) > 0 \quad (14)$$

Using the chain rule and rearranging the items, we can get the following equations:

$$I(\{x_n \mid n \in S^c\}; \theta \mid \{x_m \mid m \in S\}) - I(\{x_n \mid n \in S^c\}; \theta \mid \{x_m \mid m \in S\}, y) > 0 \quad (15)$$

$$H(\{x_n \mid n \in S^c\} \mid \{x_m \mid m \in S\}) > H(\{x_n \mid n \in S^c\} \mid \{x_m \mid m \in S\}, \theta) \quad (16)$$

The above equations indicate that effective multimodal models θ should learn a mapping across modalities, which captures information useful for explaining the dependence between $\{x_n \mid n \in S^c\}$ and the label y .

A.2 Additional Experiment Settings

Preprocessing.

Physiological signals are sampled at 125 Hz for the MIMIC-AF and MIMIC-Hyper datasets and at 100 Hz for the MESA subset. The ECG signals are band-pass filtered at $[0.5, 40]$ Hz with a Butterworth filter, while the PPG and ABP signals were band-pass filtered at $[0.5, 8]$ Hz using a Chebyshev-II filter. All signals are rescaled using min-max normalization.

Metrics.

The evaluation metrics of the AF detection tasks and blood pressure classification tasks include *Accuracy*, *Macro F1*, and *Area Under the Receiver Operating Characteristic Curve (AUC-ROC)*. We also use an additional metric *Kappa* for the sleep staging task, which measures the agreement between predictions and ground truth.

Implementation Details.

The model is trained for 100 epochs using the Adam optimizer [Kingma and Ba, 2015] with a learning rate of 0.001 and a batch size of 256. The temperature τ for contrastive learning is set to 0.07. λ_{cls} , λ_{self} , λ_{sup} , and β_1 are set to 0.25, 0.5, 0.25, 0.5, respectively. The proposed method adopts EfficientNet [Tan and Le, 2019] as the backbone to extract features from physiological signals.

Baseline.

We chose eight state-of-the-art baselines to compare against our proposed method:

- 1) Two unimodal methods aiming to address inter-subject variability in physiological signals: BPD and CSGR. BPD [Su *et al.*, 2022] uses adversarial training to disentangle task-related, subject-invariant features from label-irrelevant features to mitigate inter-subject variability. CSGR [Kim *et al.*, 2024] employs contrastive learning and deep metric learning paradigms to learn generalizable task-related features;
- 2) Three multimodal coordination methods: Multimodal SimCLR, MERL, and SiamAF. Multimodal SimCLR [Chen *et al.*, 2020] treats temporally synchronized signal segments from different modalities as positive pairs and others as negative pairs. MERL [Liu *et al.*, 2024] utilizes a contrastive learning strategy to align features within the same modality and across different modalities. SiamAF [Guo *et al.*, 2023] employs a Siamese network to encourage learning shared features across different modalities;
- 3) Two multimodal domain generalization approaches: SimMMDG and CMRF. SimMMDG [Dong *et al.*, 2023] proposes to split the learned representations into modality-specific and modality-shared parts to improve domain generalization and reconstruct the features of missing modalities during the test phase under unimodal test scenarios. CMRF [Fan *et al.*, 2024] proposes to construct modality-consistent flat loss regions via cross-modal knowledge transfer to learn generalizable multimodal representations.

In addition, we include a conventional supervised learning baseline trained on unimodal data using the standard cross-entropy loss.

For the multimodal coordination and domain generalization methods, multiple modalities are used as inputs during training, but only a single modality is used during testing. For a fair comparison with our proposed methods, EfficientNet [Tan and Le, 2019] is adopted as the backbone encoder to extract modality-specific features for each baseline method.

A.3 Additional Experiment Results

Additional Ablation Study

Table 6: Ablation study on the influence of each individual component for the MESA subset. The bold and underlined items denote the best and second-best results, respectively. The asterisk has the same meaning as Table. 2.

ID Method	MESA subset			
	Accuracy ($\uparrow$)	Macro F1 ($\uparrow$)	AUC-ROC ($\uparrow$)	Kappa ($\uparrow$)
<i>Unimodal</i>				
A1 Supervised w/ CE	0.493 $\pm$ 0.007	0.442 $\pm$ 0.005	0.720 $\pm$ 0.003	0.286 $\pm$ 0.006
A2 Supervised contrastive loss	0.508 $\pm$ 0.005	0.450 $\pm$ 0.005	0.721 $\pm$ 0.005	0.301 $\pm$ 0.006
<i>Multimodal</i>				
B1 Proposed w/o self-supervised contrastive loss	0.523 $\pm$ 0.003	0.448 $\pm$ 0.008	0.718 $\pm$ 0.004	0.305 $\pm$ 0.005
B2 Proposed w/o supervised contrastive loss	0.512 $\pm$ 0.012	<u>0.455</u> $\pm$ 0.006	0.736 $\pm$ 0.004	<u>0.306</u> $\pm$ 0.012
— Proposed PhysioGMC	<u>0.517</u> $\pm$ 0.004	0.461 * $\pm$ 0.004	<u>0.735</u> $\pm$ 0.005	0.314 * $\pm$ 0.004